

The Financial Opportunity Cost of Continued Navy Service for Primary-Care Physicians

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Purpose and Research Question

The U.S. Navy Medical Corps must understand the financial alternatives physicians face at retention decisions. Using a board-certified general internist in San Diego as the primary example, this paper asks: **How much financial compensation does a physician forgo by remaining in the Navy for the next four years rather than accepting a feasible civilian position?** The analysis informs, but does not determine, policy choices involving retention bonuses, communication of military career benefits, working conditions, and staffing models.

The issue is economically important because recruiting a physician does not ensure retention after the initial service obligation. The Government Accountability Office (GAO, 2020) found that maximum military cash compensation was below the civilian median in 21 of 27 medical and dental specialties it reviewed and recommended better civilian-wage, bonus-acceptance, and replacement-cost data. Those department-wide historical findings motivate this case study, but they do not prove a current Navy primary-care pay gap.

Economic Framework

Opportunity cost is the value of the best forgone alternative. Here, the financial opportunity cost of staying is the present value of civilian cash compensation minus Navy cash compensation over the same four years. The comparison is marginal: prior medical-school support is sunk, while future pay, additional obligations, retirement accrual, and working conditions remain relevant (Frank et al., 2022). A civilian comparator is valid only if the physician is free to separate, qualified, and reasonably able to obtain the position.

Cash compensation is only one part of the decision. Mission, camaraderie, teaching, leadership, operational medicine, assignment control, workload, and family stability can create compensating wage differentials. Pension, health coverage, tax advantages, and employer retirement contributions are financial benefits and should eventually be valued as such. The cash gap therefore measures neither total career value nor the bonus necessary to retain a physician.

Method and Assumptions

The base case is an O-4 general internist just past 10 years of active service, with dependents, assigned to ZIP code 92134. The horizon is 2027-2030 in constant 2026 dollars. Rank remains O-4; basic pay advances from the over-10 rate in years 1-2 to the over-12 rate in years 3-4. A 5% real discount rate and year-end payments are used. The civilian benchmark is Marit Health's published San Diego internist average of \$459,057, held constant in real terms. Because it is a mixed-employer, self-reported average rather than a matched offer, results are conditional on its comparability (Marit Health, 2026).

Compensation Inputs

Component	2027-2028	2029-2030
Basic pay	\$113,040	\$118,660
BAH with dependents	\$60,984	\$60,984
BAS	\$3,942	\$3,942
Incentive pay	\$43,000	\$43,000
Board-certification pay	\$8,000	\$8,000
Assumed 4-year retention bonus	\$48,000	\$48,000
Total Navy gross cash	\$276,966	\$282,585

Table 1. Itemized base-case cash compensation. Basic pay, BAS, and special-pay inputs follow published FY26 tables; the San Diego BAH input requires an individual entitlement check. The \$48,000 retention bonus is conditional on Navy eligibility and an executed agreement, not a verified individual entitlement (DFAS, 2026a-c; Defense Travel Management Office, n.d.).

The model discounts each annual difference using: $\text{discounted gap}_t = (\text{civilian cash}_t - \text{Navy cash}_t) / (1.05)^t$. BAH and BAS receive federal income-tax exclusions, so equal gross compensation would not imply equal take-home pay (Internal Revenue Service, 2025). A complete comparison would apply consistent household tax assumptions and add differences in health coverage, retirement contributions, malpractice costs, and transition expenses.

Pension Eligibility and the Missing Dynamic Incentive

The base case begins at 10 years and ends at 14, before the usual 20-year active-duty retirement threshold. No pension payment is therefore included in the four-year cash table. This is a major limitation: the model does not clearly quantify how the incentive to remain increases as a physician approaches 20 years. Leaving at 10 years means giving up a distant and uncertain pension; leaving at 18 or 19 years means forfeiting a valuable lifetime benefit after only one or two more years of service. The opportunity cost of separation therefore rises nonlinearly as eligibility approaches - the familiar 'golden handcuffs' effect.

A simplified sensitivity illustrates the mechanism. Assuming a \$120,000 High-3 basic-pay base, 30 years of constant real pension payments, and a 5% discount rate, a BRS pension of \$48,000 annually has a conditional present value of about \$453,000 at 10 years, \$669,000 at 18 years, and \$703,000 at 19 years. These are not amounts to subtract directly from the four-year cash gap: they assume the member reaches 20 years and omit foregone civilian earnings, taxes, uncertainty, and alternative Reserve benefits. They demonstrate why career stage must be modeled explicitly (Department of Defense, n.d.).

Moonlighting also requires a separate scenario because it adds labor and may be limited by deployments, reassignment, approval, fatigue, and malpractice expenses. Educational benefits already received under HPSP, HSCP, or USUHS are sunk and should not be counted again as new retention compensation.

Results

Year	Navy cash	Civilian benchmark	Annual gap	PV gap (5%)
2027	\$276,966	\$459,057	\$182,091	\$173,420
2028	\$276,966	\$459,057	\$182,091	\$165,162
2029	\$282,585	\$459,057	\$176,472	\$152,443
2030	\$282,585	\$459,057	\$176,472	\$145,184
Total	\$1,119,102	\$1,836,228	\$717,126	\$636,209

Table 2. Four-year gross-cash comparison. Positive gaps favor the civilian benchmark. Present values use unrounded inputs.

The base case produces a four-year nominal cash gap of \$717,126 and a present-value gap of approximately **\$636,209**. The over-12 longevity increase narrows the annual gap beginning in 2029 but does not reverse it. At a 3% real discount rate, the gap is \$666,715; at 7%, it is \$607,907. A \$10,000 change in annual civilian compensation changes the four-year present value by about \$35,460. If the assumed \$48,000 annual retention agreement is unavailable, the four-year present-value gap increases by about \$170,206.

These results are accounting estimates, not predictions of separation. They do not show how strongly physicians respond to pay, the value physicians place on noncash job attributes, or the cost to the Navy of replacing a departing physician. The mixed-employer benchmark may overstate or understate the best feasible alternative for a particular officer. A matched offer would need to account for workload, call, experience, practice setting, location, and employer-funded benefits.

Policy Options

The initial recommendation is a limited, testable communication campaign aimed at physicians approaching retention decisions. Existing channels could explain real opportunities for mission-focused practice, leadership, teaching, research, and operational medicine while separately clarifying special-pay eligibility, tax treatment, and retirement consequences. Communication may correct an information gap, but it cannot close the cash gap or make inaccessible opportunities available. The campaign should therefore be candid about deployments, geographic mobility, administrative burden, and assignment uncertainty.

Success should be measured through baseline awareness, access to advertised opportunities, agreement acceptance, and later retention compared with standard communication. Clicks and views are not retention outcomes. If awareness increases without changes in acceptance or retention, the Navy should redirect attention toward compensation or working conditions. Army research suggests that family well-being, deployment effects, workload, and workplace factors matter alongside pay, although those findings should not be treated as Navy-specific causal estimates (Vie et al., 2024; Wojcik et al., 2020).

Bonus advocacy is a complementary option. A larger bonus directly changes financial compensation, but the cash gap alone cannot identify the optimal amount. Evaluation must estimate the marginal cost per additional physician retained, including payments to physicians who would have stayed without the increase. Agreement uptake must also be separated from genuinely additional service obtained.

Policy Options, Continued

A third option is to shift appropriate primary-care work toward nurse practitioners and physician assistants while preserving physician capacity for complex care, teaching, supervision, and operational readiness. VA studies demonstrate that advanced practice clinicians can manage substantial primary-care workloads and, in selected populations, may be associated with similar measured outcomes and lower utilization or costs (Liu et al., 2020; Morgan et al., 2019). These findings do not establish one-for-one substitution or a Navy savings rate. A Navy pilot should compare total team costs, access, quality, downstream utilization, workload, and readiness under different staffing configurations.

Working-condition reforms may be equally important. Military and civilian studies link burnout and turnover-related outcomes with administrative burden, workload, schedule control, leadership support, and work-life concerns (Agarwal et al., 2020; Korona-Bailey et al., 2024; Wilk et al., 2023). Advertising attractive missions without addressing these barriers may undermine credibility. Compensation, job design, and communication should therefore be evaluated as related but distinct interventions.

Limitations

The analysis uses one O-4 profile, one location, and one civilian benchmark. It assumes continued eligibility for a four-year retention agreement, holds compensation constant in real terms, and excludes promotion. It also omits taxes, health coverage, employer retirement contributions, malpractice and transition costs, and the probability-adjusted value of future retirement benefits. These omissions may narrow or widen the measured gap.

Most importantly, the fixed four-year horizon masks the changing incentive created by pension proximity. The cash model should not be interpreted as though a physician at 10 years faces the same stay-or-leave decision as one at 18 or 19 years. Future work should model career-stage scenarios through retirement eligibility, apply the correct High-3 or BRS rules, distinguish vested TSP balances from the defined-benefit pension, and compare active-duty, Reserve, and civilian paths using consistent timing and discounting.

The model also does not measure actual Navy primary-care attrition, causal readiness losses, retention elasticities, or replacement cost. Before using the estimate for policy advocacy, the Medical Corps should validate individual pay entitlements and comparable civilian offers by specialty and career stage. Any intervention should track eligible populations, costs, additional service obtained, working conditions, and actual retention using a credible comparison group.

Conclusion

For the modeled O-4 general internist, remaining in the Navy rather than taking the published San Diego benchmark produces a four-year gross-cash opportunity cost of approximately \$636,209 in present value at 5%. The estimate is substantial but incomplete. It excludes tax and benefit differences and does not fully capture the rapidly increasing value of staying as pension eligibility approaches.

The best near-term response is not to assume that one intervention solves retention. The Navy should test accurate, targeted communication as a low-cost first step; evaluate larger bonuses using marginal cost per additional physician retained; improve working conditions where operationally feasible; and pilot team-based staffing for appropriate clinical work. Better specialty- and career-stage data are necessary to determine which combination produces the greatest retention and readiness value.

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